

The characterization of porcine retinal explants and their use as an ex vivo model for retinal gene therapy

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Purpose: Inherited retinal diseases (IRD) significantly contribute to vision loss worldwide,

impacting the quality of life for affected individuals. Gene editing (GE) offers a promising

future solution as an effective treatment through restoration of gene function, but translation into the clinic has proven difficult. Therefore, effective models to evaluate the

specificity and efficacy of therapeutic GE are of particular importance.

Methods: Since findings from our preliminary in vitro experiments failed to translate to an

in vivo pig model, porcine retinal explants (RE) were selected as an alternative tool for cellular systems. RE were analyzed at 5 different time points, days 0, 1, 3, 7, and 14, to assess their stability in culture. Characterization of RE was carried out by

immunofluorescent staining, qPCR, and bulk RNAseq. Once established, RE were treated

with Adeno-associated virus (AAV)9 in combination with a GFP reporter at different application sites and their transduction into retinal cells was assessed by flow cytometry

and immunofluorescent staining.

Results: Histological analysis showed that RE retained their structure well with minimal

signs of degeneration and were able to rapidly adapt and remain stable under culture

conditions. However, at the molecular level, stress-related processes accompany adaptation to culture, highlighting the need for rapid culture after isolation and immediate

treatment with vectors. By applying AAV9 with GFP at different application sites, we also

found potential shifts in GFP expression in retinal cell populations dependent on RE

orientation. Finally, through flow cytometry analyses, we were further able to discriminate

rod cells and assess GFP expression within a specific cellular population.

Conclusion: The conflicting results previously observed moving from an in vitro to in vivo

model highlight the need for a better predictive tool which more closely replicates the

post-mitotic tissues of the retina when assessing GE techniques. RE are an ideal candidate to bridge this gap since they preserve tissue architecture, physiological relevance, and have already demonstrated value as a model for GE vector testing.

Keywords: Gene therapy, Porcine retinal explants, Adeno-associated virus, Inherited retinal diseases